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6 February 1973
SUPERSEDING
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27 April 1972

MILITARY SPECIFICATION

MODIFICATION KIT, LIQUID FUEL
FILTER-SEPARATOR KMU-417/F

1. SCOPE

1.1 This specification covers a liquid fuel filter-separator modification kit, designated KMU-417/F.

2. APPLICABLE DOCUMENTS

* 2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

L-P-378	Plastic Sheet and Strip, Thin Gauge, Polyolefin
PPP-B-566	Box, Folding, Paperboard
PPP-B-601	Box, Wood, Cleated-Plywood
PPP-B-621	Box, Wood, Nailed and Lock Corner
PPP-B-636	Box, Fiberboard
PPP-B-640	Box, Fiberboard, Corrugated, Triple-Wall
PPP-B-676	Box, Setup

Military

MIL-P-116	Preservation, Methods of
MIL-F-8901	Filter-Separators, Aviation and Motor Fuel, Ground and Shipboard Use, General Requirements and Test Procedures for
MIL-F-52308	Filter Element, Fluid Pressure

STANDARD

Military

MIL-STD-129	Marking for Shipment and Storage

FSC 4330

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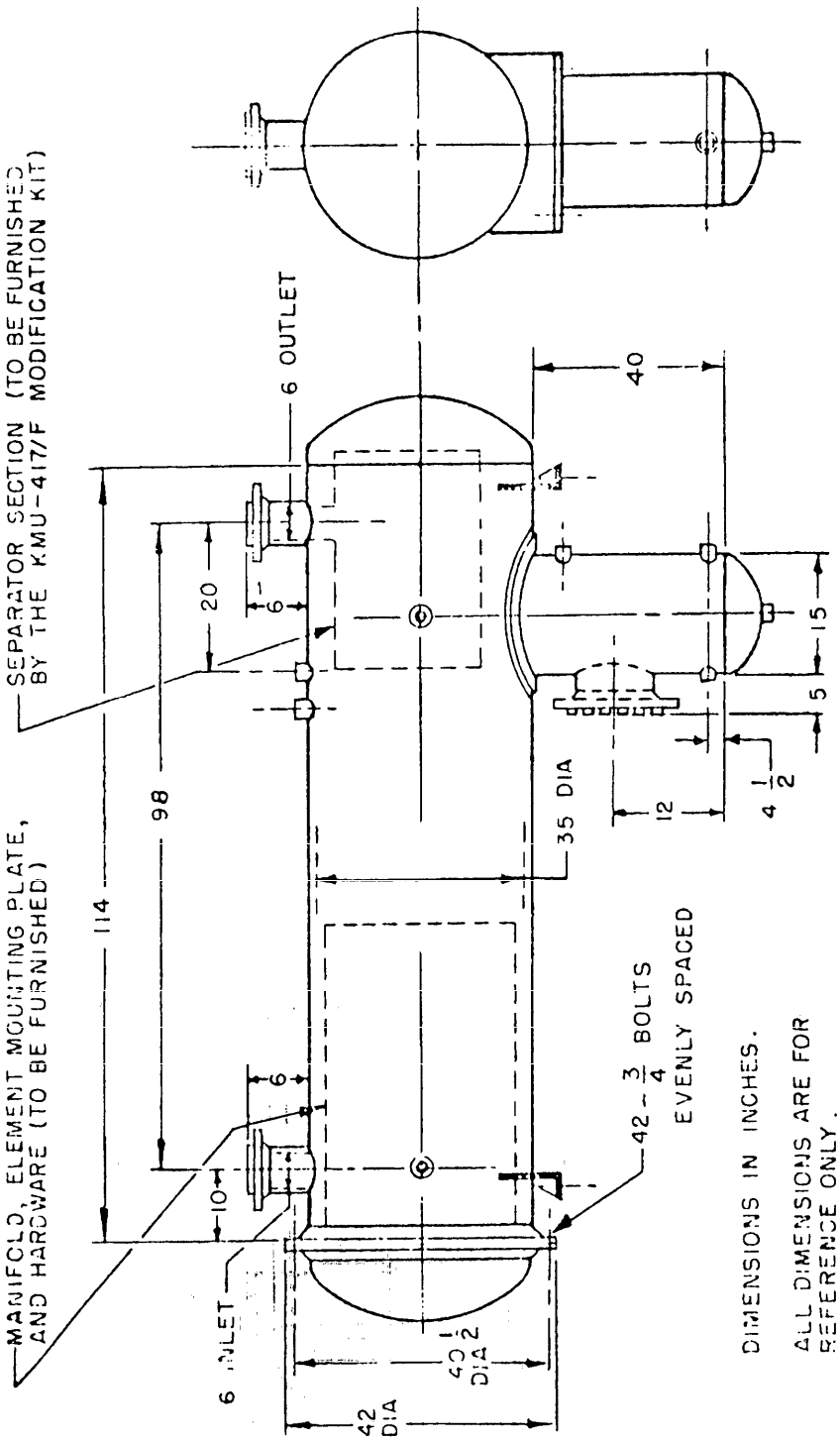


*FIGURE 1. Filter-Separator Housing

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* 3.3.1.2 Filter elements (coalescers). The modification kit shall be designed to accept 30 filter elements conforming to MIL-F-52308. These elements may be either single or double stacked.

* 3.3.1.3 Piping manifold. The inlet manifold for the filter (coalescer) stage and the outlet manifold for the separator stage shall be provided with a ring flange and shall be designed to be installed through the inlet and outlet nozzles in the filter separator vessel. The ring flanges shall be sealed in between the flange of the pressure vessel and the flange on the fuel supply piping. Additional flange bolts, seals, nuts, and lockwashers shall be included with the modification kit, as required.

* 3.3.1.4 Filter (coalescer) element mounting plate. A filter element mounting plate designed to accept 30 elements conforming to MIL-F-52308 shall be provided. It shall be equipped with either 15 or 30 element mounting tubes, depending upon whether the elements will be single or double stacked. The tubes shall be welded or permanently sealed to the mounting plate and shall be sized to accept and to seal the elements. The mounting plate shall be provided with a minimum of two handles spaced to facilitate its removal either manually or with the use of a hand hoist. The mounting plate shall be designed for installation in both filter-separator vessels shown on figures 1 and 2. To provide the installation capability in both size vessels, each mounting plate shall be equipped with a removable extension ring flange. The flange would be installed and used in the larger diameter vessel, but would not be required for the smaller diameter vessel. The removable ring flange shall be designed to be installed to the mounting plate by the use of common handtools. The mounting plate, with the ring flange installed, shall withstand a differential pressure of 75 psi without leakage. Element end seals, intermediate element seal plates, element holding rods, supports, and other hardware shall be provided as required. Consideration shall be given in the design of the main element mounting plate to provide a manhole and cover plate to permit access to the elements during installation. The use of the manhole would facilitate removal, installation, and securing of the elements without removal of the element mounting plate every time elements are required to be changed or inspected. Filter elements may also be mounted on the manhole cover plate if such a design is used.

3.4 Performance. The modification kit shall be capable of the following:

a. Being installed in both housings shown on figures 1 and 2

* b. Meeting the 5 percent water removal requirements specified in MIL-F-8901.

* c. Meeting the red iron oxide dry test specified in MIL-F-8901.

3.5 Manufacturer's part number. The manufacturer's part number and drawing number shall be the same.

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* 4.3.2 Preproduction tests. The preproduction tests shall consist of the tests specified herein and the following tests as specified in MIL-F-8901:

a. Examination of product

b. Five percent water removal

c. Red Iron Oxide (dry).

4.4 Acceptance tests. The acceptance tests shall consist of the individual tests.

4.4.1 Individual tests. Each modification kit shall be subjected to the test specified in 4.5.1. Each separator element shall be subjected to the test specified in 4.5.2.

4.5 Test methods

4.5.1 Examination of product. The modification kit shall be examined to determine compliance with the requirements specified herein with respect to materials, workmanship, dimensions, and markings.

4.5.2 Separator inspections. The separator element shall be subjected to an inspection to determine that all joints are properly sealed and the wire mesh is properly attached to insure integrity. The separator element shall be mounted in a fixture that will allow the element to be rotated by hand 360 degrees in a level plane. At least 1/2 inch of water shall be placed inside the nonfuel-wetted canister. The element shall be rotated 360 degrees and the separator element inspected for leakage. Leaks shall indicate defects in the hydrophobic medium (teflon screen) and shall be cause for rejection.

4.5.3 Flow rate. During the full-scale test conditions, the flow rate capacity for the modification kit shall be 600 gpm.

* 4.5.4 Field installation test. Unless otherwise specified (see 6.2), a preproduction modification kit shall, prior to production approval, be subjected to a field installation test. A preproduction modification kit shall be shipped to each Air Force base identified for delivery of other production items. Upon arrival of the equipment, an installation test shall be performed. During this test, mechanical clearances shall be noted and assurances obtained that the modification kit can satisfactorily be installed without difficulty. Also, that elements can be removed when required and the kit can satisfactorily be installed in like filter-separators scheduled to be modified. (See 6.3.) A preproduction modification kit drawing shall be mailed to each Air Force base identified for delivery of other production items. Approval of the drawing shall be made by the Base Civil Engineer prior to production approval.

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style optional) or in a class 2, grade A, style E container conforming to PPP-B-640. Closure shall be in accordance with the applicable container specification.

5.2.3 Level C. Modification kits, preserved and packaged as specified in 5.1, which require overpacking for acceptance by the carrier, shall be packed in shipping containers which will ensure safe delivery at the lowest transportation rate. Containers shall meet Consolidated Freight Rules or regulations of other common carriers as applicable to the mode of transportation.

5.3 Marking. Unit packs and exterior shipping containers shall be marked in accordance with MIL-STD-129. The nomenclature shall be:

MODIFICATION KIT, LIQUID FUEL
FILTER-SEPARATOR KMU-417/F

6. NOTES

* 6.1 Intended use. The KMU-417/F modification kit is intended to upgrade the design of the horizontal filter-separators installed in fixed hydrant fueling systems. The modification kit has application in the following model filter-separators:

Briggs-Bowser-Keene

BFS-50-600-V31
BFS-50-600

Erie Company
(600 GPM)

4882-50
4872-50
4876-50
4880-50

Fram Corporation
(Warner-Lewis)
(600 GPM)

HP-600-ANC
HPB-600-ANC
HPQ-600-ANC
HPBQ-600-ANC
FEP 500-ANC
FEPB-500-ANC
FEPBQ-500-ANC
FE-10E

HP-600-ANCH
HPB-600-ANCH
HPQ-600-ANC
HPBQ-600-ANC
FEP-500-ANCH
FEPB-500-ANCH
FEPQ-500-ANCH

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

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MATERIAL PROCURED UNDER A

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A. GIVE PARAGRAPH NUMBER AND WORDING.

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